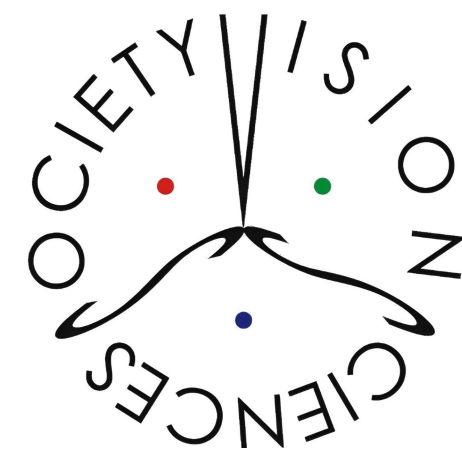


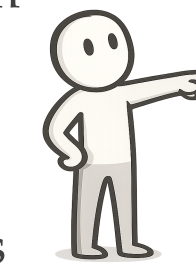


I'll Believe It Unless It's Too Absurd: Spontaneous Visual Perspective-Taking as Prior Based Heuristic Inference

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*An extended version of this paper has been accepted for publication in *Cognition*. See full details

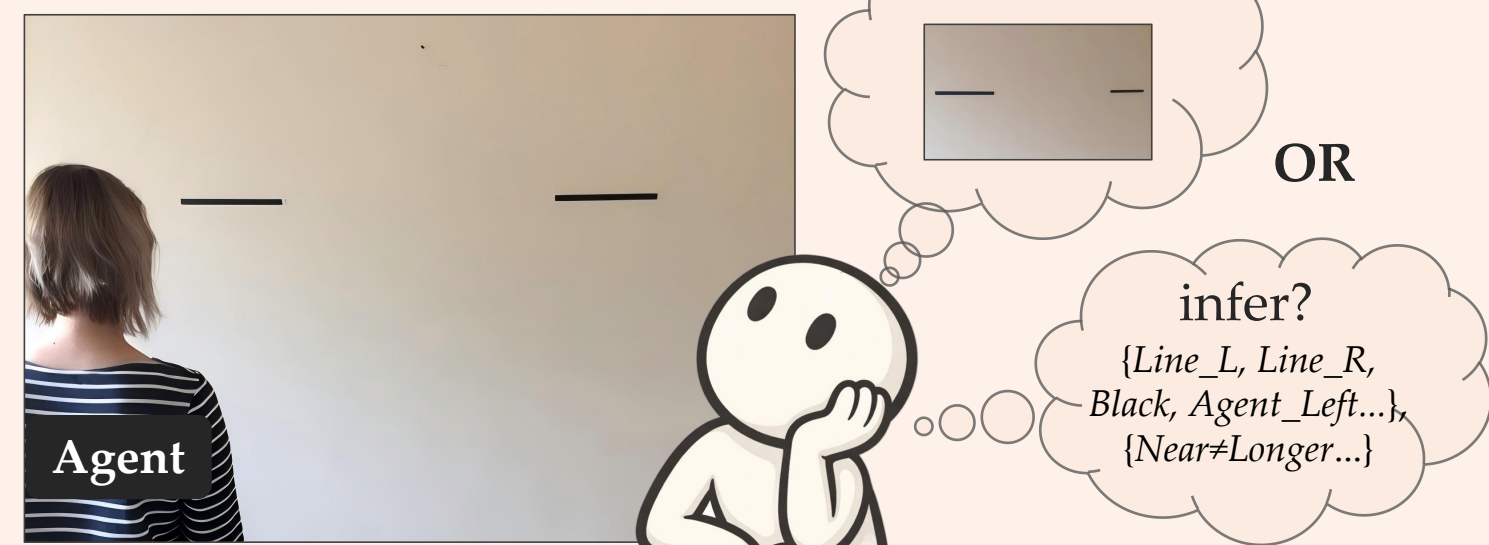


View
PDF

View
Poster

Core Question

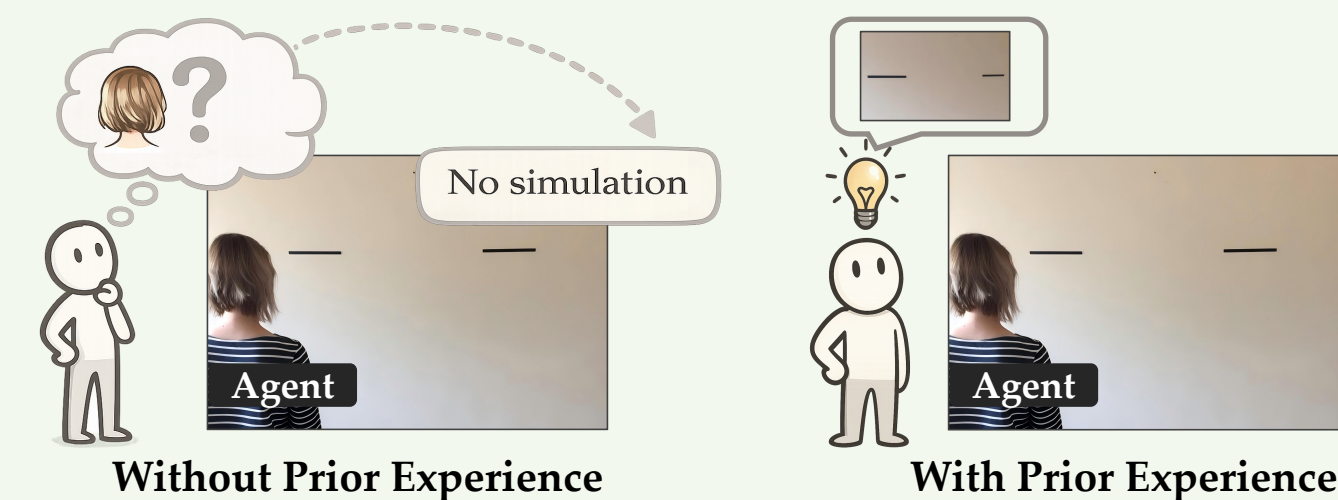
How do we take others' visual perspectives?



- Do we **reconstruct** what others actually see (*perceptual simulation*)? Or do we **infer** it using *heuristic rules* about vision?
- **Explicit VPT** favors *heuristic* accounts — but **implicit VPT** remains unresolved.

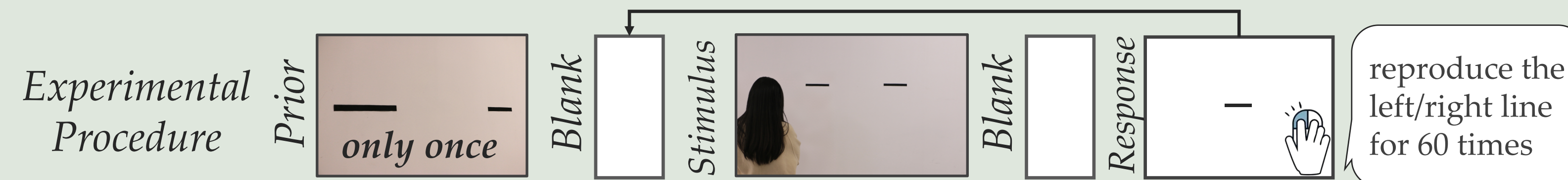
Key Hypothesis

But why does simulation seem to fail?

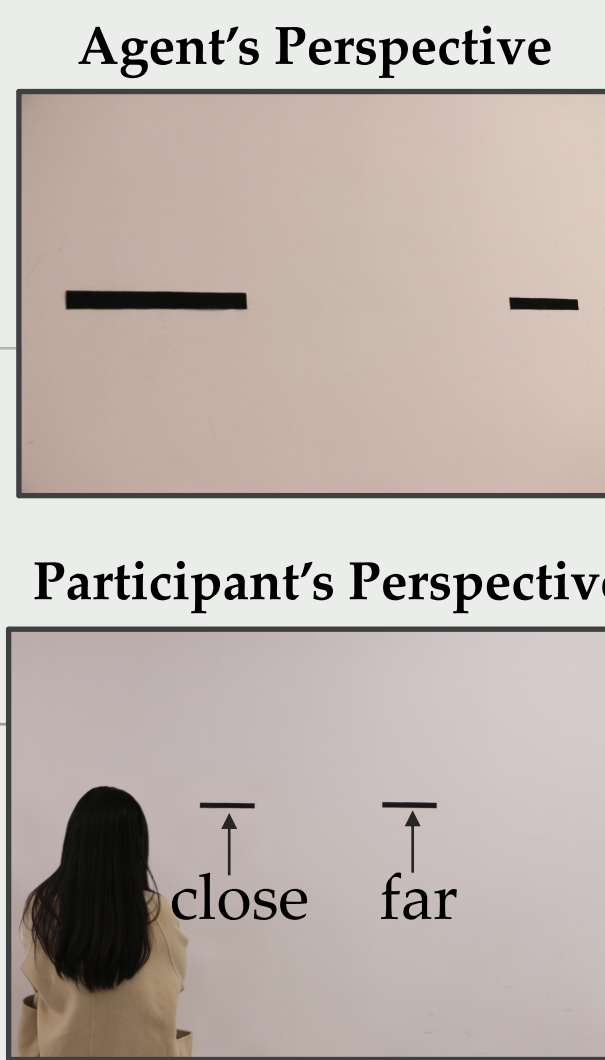
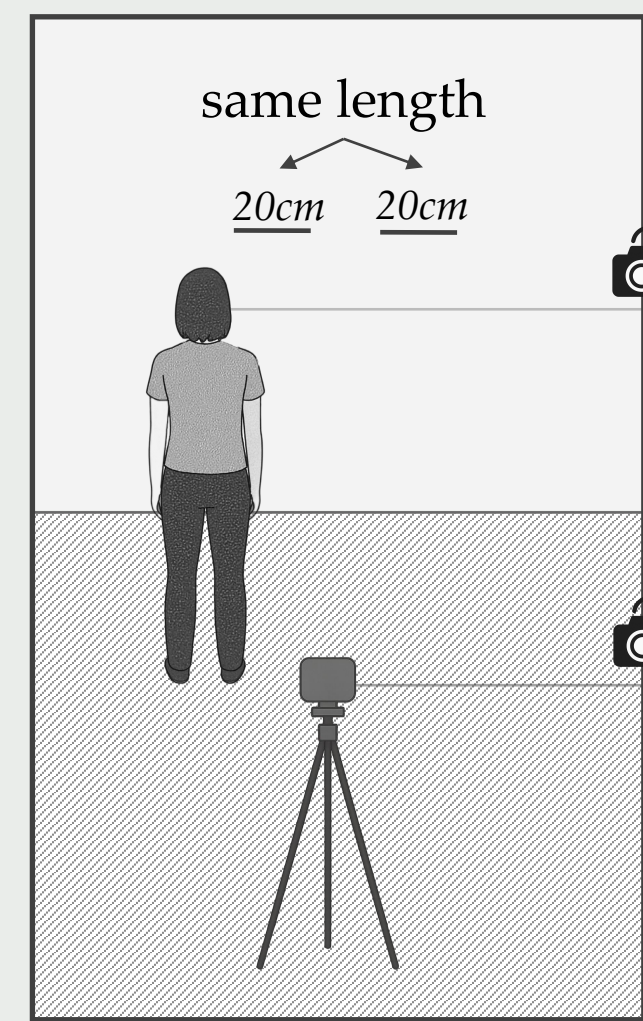


- Simulation may not fail because it can't, but the **agent's perspective is unknown**.
- **Test different priors in implicit VPT:**
 - *Perceptual Simulation Account* predicts bias only with correct priors.
 - *Heuristic inference* predicts bias even with incorrect priors.

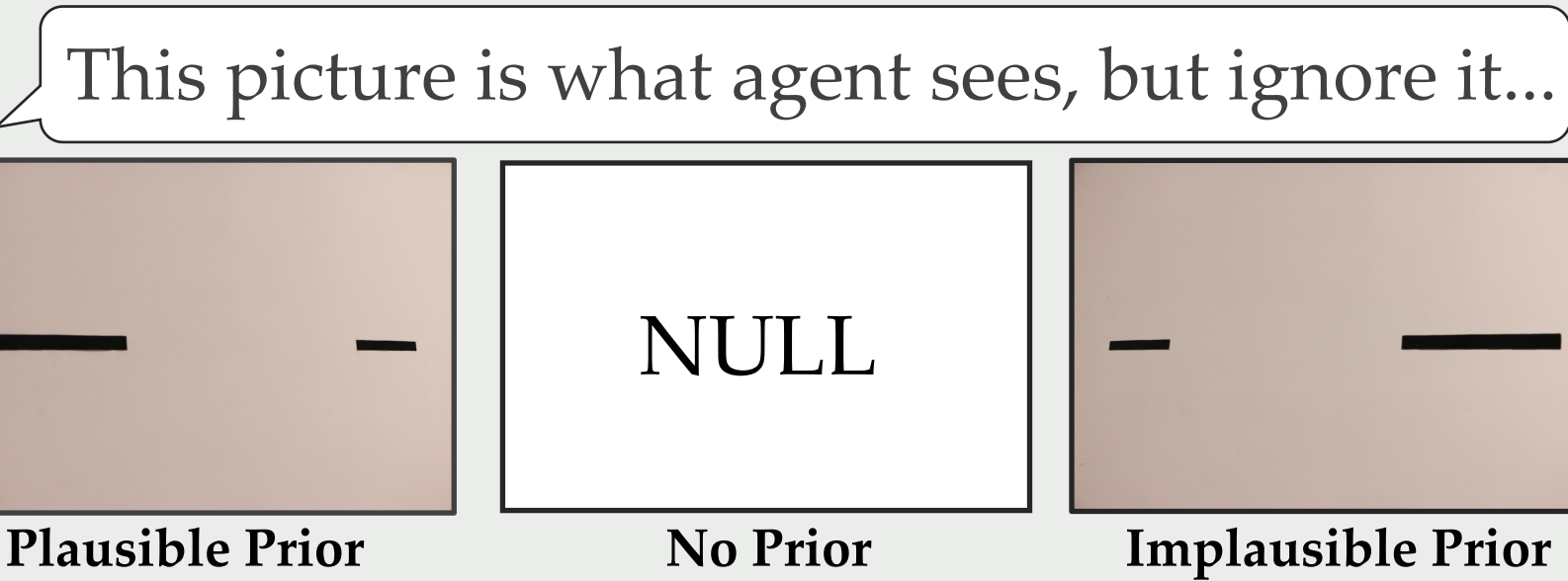
Design



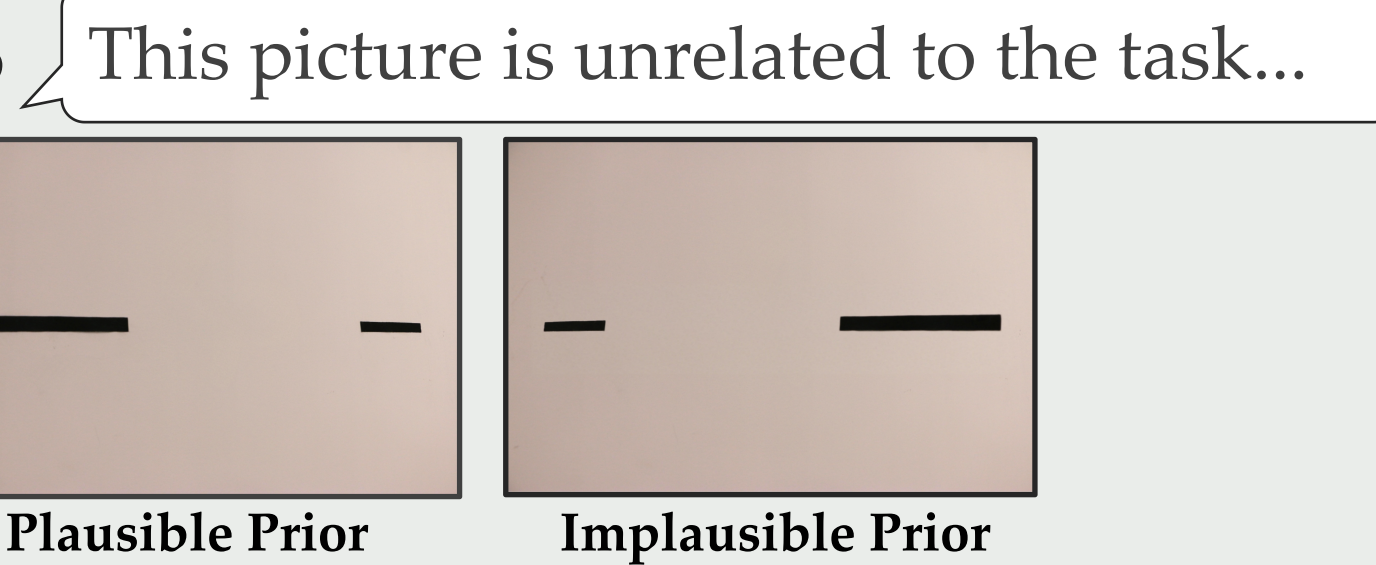
•Exp1



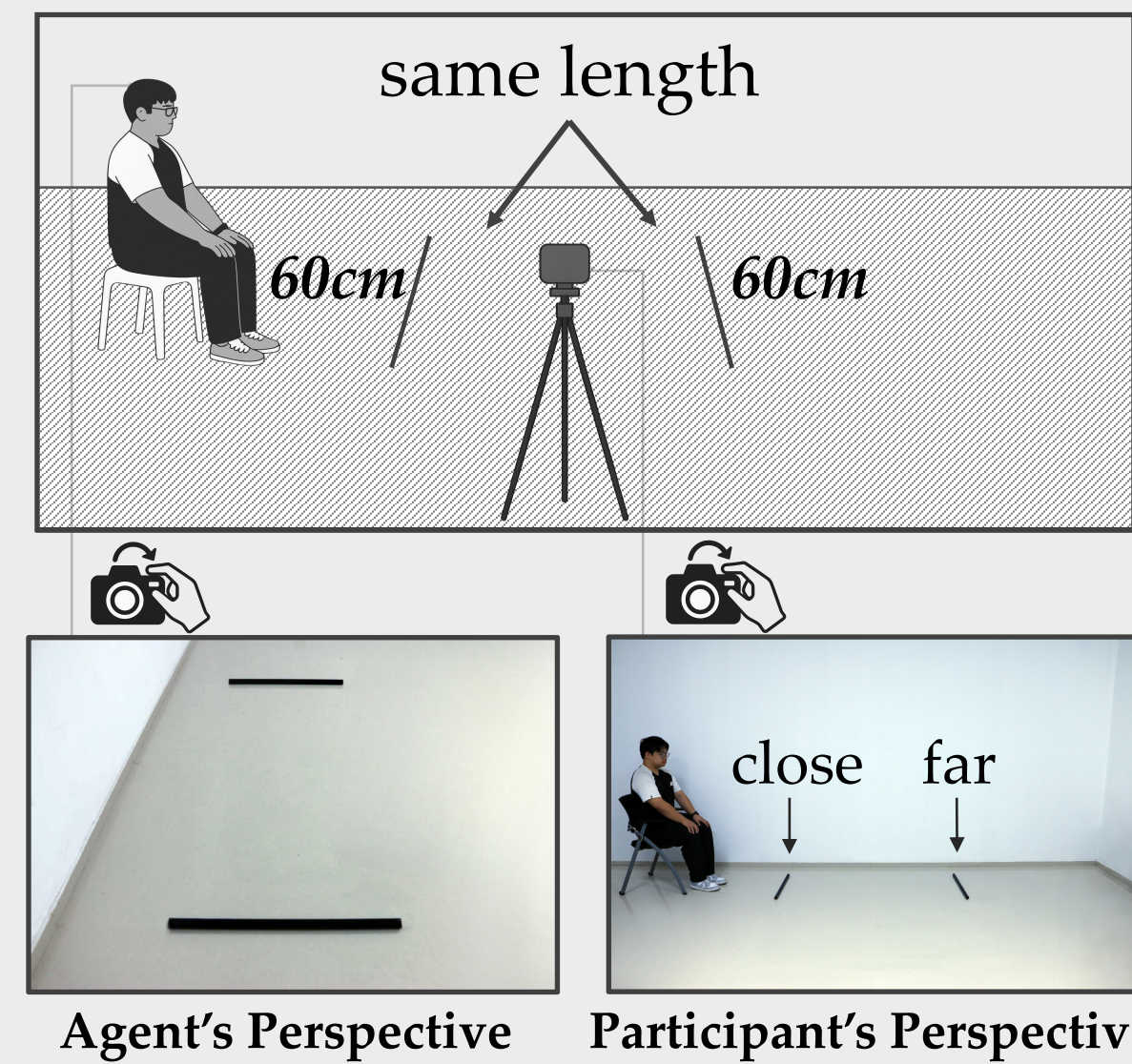
•1a



•1b



•Exp2



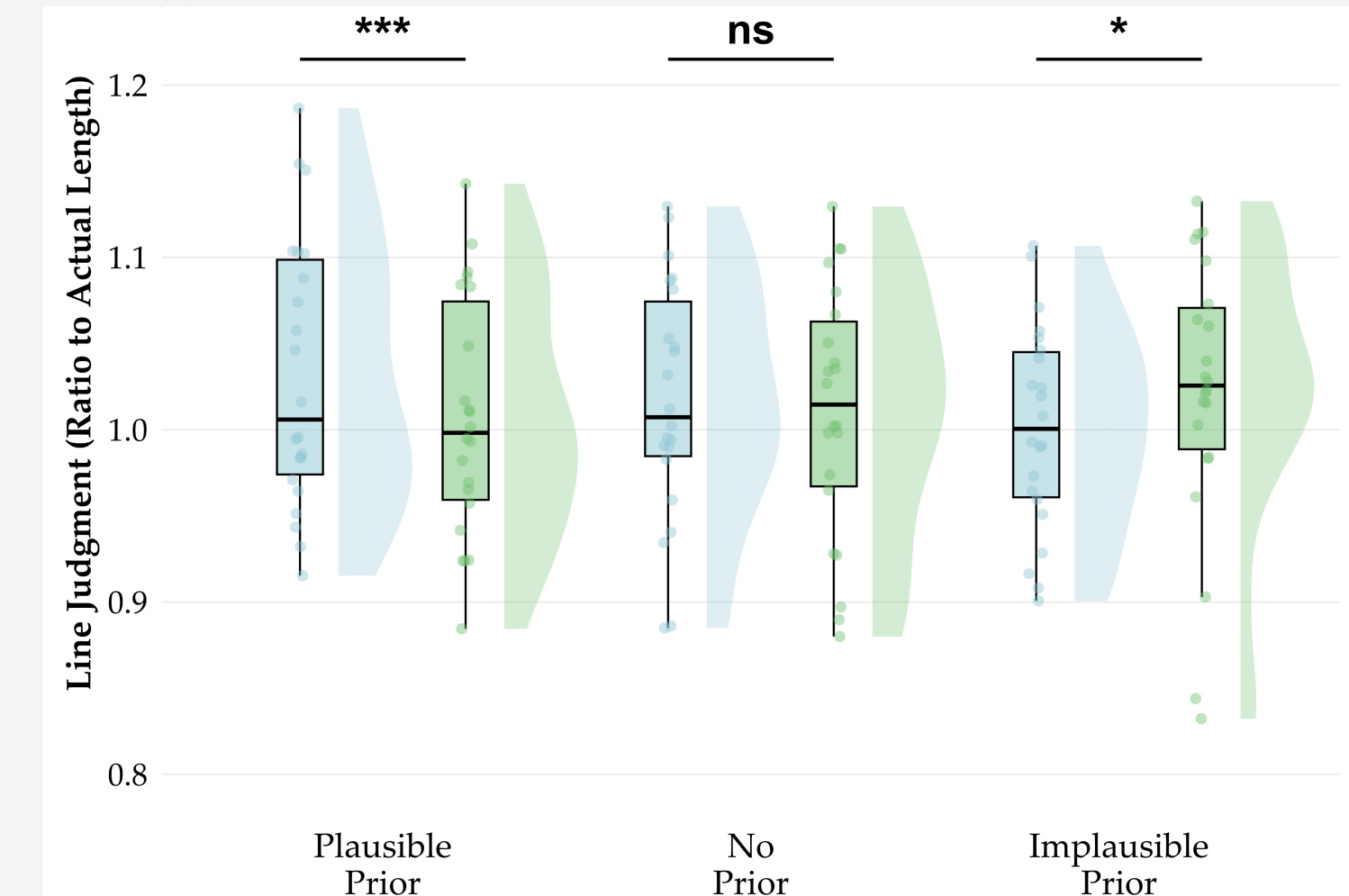
Prior: Plausible Prior, No Prior, Implausible Prior

• Priors vary in plausibility

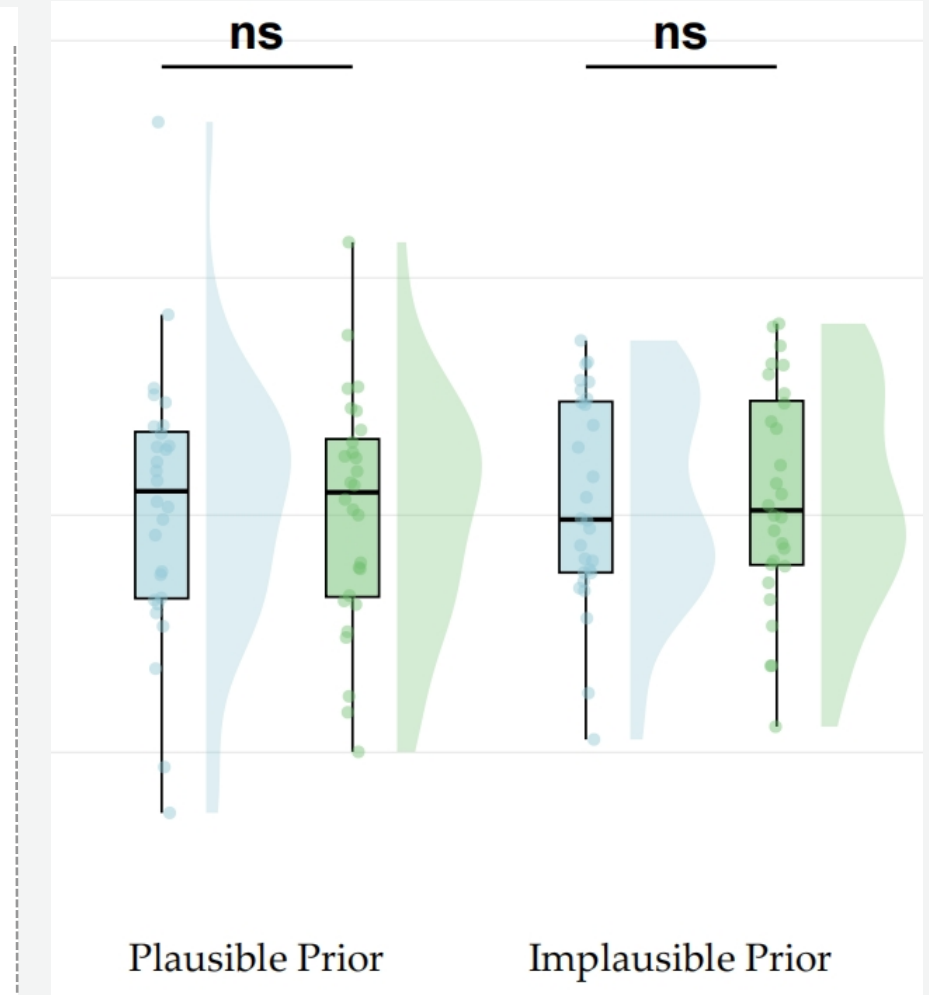
Instructions (same as Exp 1a):
“The picture shows the **agent's perspective** in the next scene. But please **ignore** it and **focus only on remembering the line lengths** in the upcoming image.”

Results

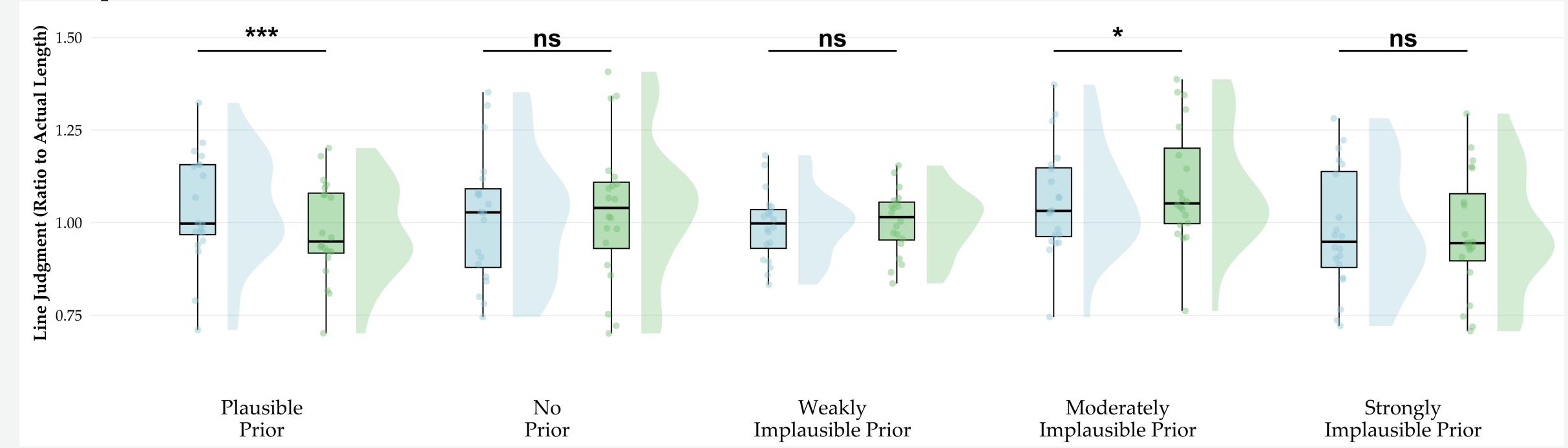
•Exp1a



•Exp1b



•Exp2



Conclusion

- **Implicit VPT** remains best explained by *heuristic inference* rather than *perceptual simulation*.
- Judgments are biased by both correct and incorrect priors (Exp. 1a), and this bias is not driven by priming (Exp. 1b), ruling out simulation-based reconstruction.
- Heuristic inference is plausibility-weighted, integrating prior information with self-viewed input, while extreme priors are discounted (Exp. 2).